The universal property of the free group: group homomorphisms out of F(A) are the functions from A

The Univalent Foundations Program

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Editorial note. The counted theorem carries no \label in the pinned source; its identity is fixed by the pinned file plus the original number 6.11.6, obtained from the label-number dump and by counting the theorem-class environments of section 6.11 of the same file (the neighbouring labelled results match: thm:free-monoid is 6.11.5 and thm:freegroup-nonset is 6.11.8).

Editorial note. The proof environment begins on the line immediately after \end{thm}, so the pairing is by adjacency, and the retained proof range 1620-1625 is the whole environment.

Editorial note. The retained context is the author's own paragraph reading off the constructors of F(A) and concluding that it is a group, which the statement presupposes.

Editorial note. The retained counter values (chapter 6, section 11, theorem offset 5) make the retained theorem print as Theorem 6.11.6, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: the universal property of the free group, Theorem 6.11.6.

Allowed standard prerequisites (editorial).

- The free group as a higher inductive type. F(A) is generated by eta : A -> F(A), a multiplication, an identity, an inverse and the group axioms as path constructors, together with a 0-truncation constructor.. Its recursion principle is the entire forward direction of the proof at source line 1621, and its computation rules give the homomorphism property at line 1622.
- Recursion and induction principles of a 0-truncated higher inductive type. A map F(A) -> G into a set G is determined by its values on the constructors, and an equality between two such maps is determined on the generators.. The right inverse is obtained by recursion at line 1621 and the left inverse by induction at line 1624.

- Group homomorphisms. hom_Group(F(A), G) is the set of group homomorphisms from F(A) to G.. The statement is phrased with that set, and the proof must show that the map obtained from f is a homomorphism, which it gets from the computation rules at line 1622.

The first constructor says that A maps to $F(A)$. The next three give $F(A)$ the operations of a group: multiplication, an identity element, and inversion. The three constructors after that assert the axioms of a group: associativity, unitality, and inverses. Finally, the last constructor asserts that $F(A)$ is a set.

Therefore, $F(A)$ is a group. It is also straightforward to prove:

Theorem 6.11.6. *$F(A)$ is the free group on A. In other words, for any (set) group G, composition with $\eta : A \to F(A)$ determines an equivalence*

$$\hom_{\mathrm{Group}}(F(A), G) \simeq (A \to G)$$

where $\hom_{\mathrm{Group}}(-, -)$ denotes the set of group homomorphisms between two groups.

Proof. The recursion principle of the higher inductive type $F(A)$ says *precisely* that if G is a group and we have $f : A \to G$, then we have $\bar{f} : F(A) \to G$. Its computation rules say that $\bar{f} \circ \eta \equiv f$, and that $\bar{f}$ is a group homomorphism. Thus, $(- \circ \eta) : \hom_{\mathrm{Group}}(F(A), G) \to (A \to G)$ has a right inverse. It is straightforward to use the induction principle of $F(A)$ to show that this is also a left inverse. □